AMAZON ML CHALLENGE 2026

Business Entity Resolution

Determine which business records, arriving from three independent sources with no shared identifier, describe the same real-world business.

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THE PROBLEM

From sign-up to matched records



A business signs up. We capture its **name** and **address**.

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THE PROBLEM

From sign-up to matched records

SOURCE 1
reference, deduplicated

NAME
Acme Robotics Inc.

ADDRESS
500 Market St,
San Jose

SOURCE 2
vendor A format

NAME
Acme Robotics
Incorporated

ADDRESS
500 Market Street,
San Jose CA

SOURCE 3
vendor B format

NAME
Acme Robotics

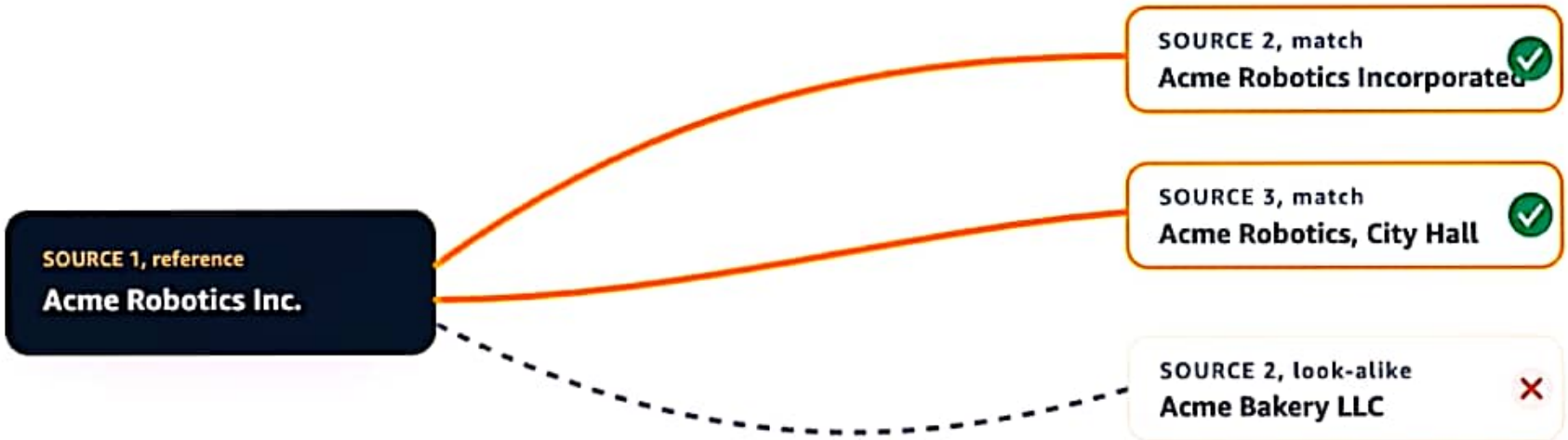
ADDRESS
Nr. City Hall,
San Jose

One business, three representations, each from a different data vendor, with **no shared ID**.

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THE PROBLEM

From sign-up to matched records

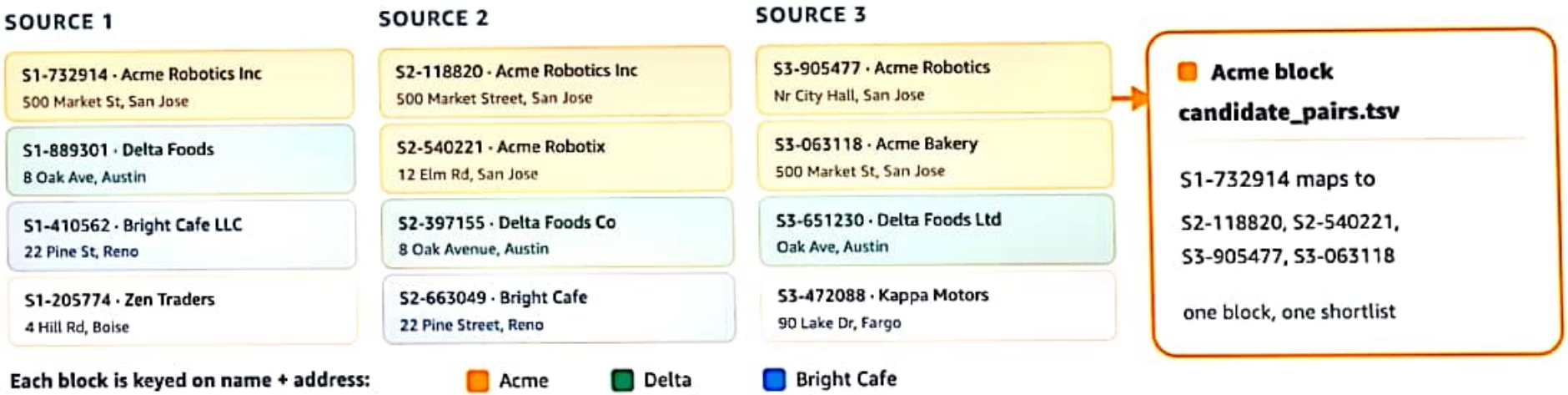


Entity resolution links every Source 1 entity to its matches: zero, one, or many.

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THE PROBLEM

From sign-up to matched records

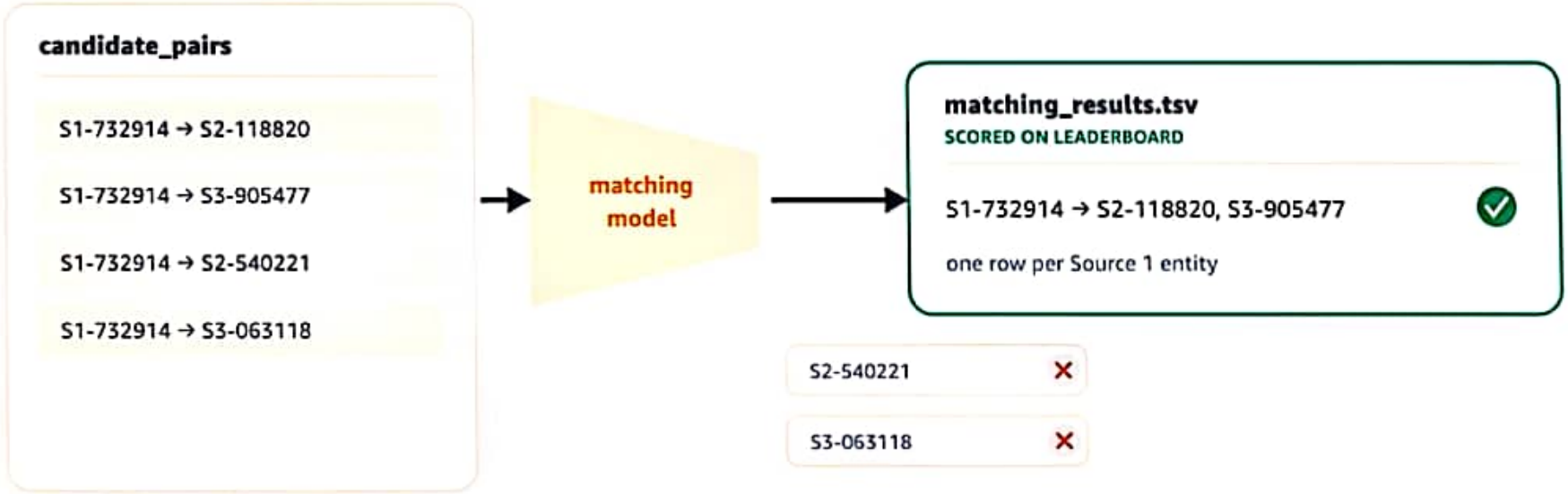


Blocking buckets records by a cheap key built from **name and address**, so similar names or shared addresses group together, matches and look-alikes alike.

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THE PROBLEM

From sign-up to matched records



The model filters candidates and keeps only true matches, producing matching_results.tsv.

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THE DATASET

Train to learn, test to prove it

- **Training set**
  - Records across all **3 sources**
  - **Ground-truth labels:** `train_ground_truth.tsv`
  - Use it to build and evaluate your model

- **Test set**
  - Same **3 sources, no labels**
  - Produce a prediction for every Source 1 entity
  - This is what the leaderboard scores

Ground truth is **one row per Source 1 entity**: its id maps to a comma-separated list of all matching Source 2 and Source 3 ids, so a single row holds the entity's full match set (empty when it matches nothing). All files are **tab-separated** (`.tsv`); read with `sep="\t"` .

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WHAT YOU SUBMIT

A file, then a package

- **During the challenge**
  - Upload `matching_results.tsv`
  - **The only file scored** on the leaderboard
  - One row per Source 1 entity, empty list means no match
- **Final package (.zip)**
  - `output/matching_results.tsv`, final matches
  - `output/candidate_pairs.tsv`, blocking set (audited)
  - `code/…`, runnable pipeline
  - `Documentation_template.md`, methodology

> Run `utils/validate_submission.py` to catch any formatting errors before you submit.

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JUDGING & TIPS

Scored on macro $F_{0.5}$, precision-weighted

$$F_{0.5} = (1.25 \times \text{Precision} \times \text{Recall}) / (0.25 \times \text{Precision} + \text{Recall})$$

- A **false merge** costs about twice as much as a miss, so when unsure, do not merge.
- **Singletons** are Source 1 entities with no match in Source 2 or 3. Predict an empty list for them and you earn a full 1.0; predict any match and you score 0.
- **Blocking sets your recall ceiling**, so invest there first. You cannot match a record you never consider.
- Account for **region-specific patterns** in both names and addresses.

> ⚠ **No external data lookup.** External databases, APIs, and geocoding services are strictly prohibited. Use only the provided data.

Type here to search
34°C Sunny
ENG
10:54 AM
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All the best

Build something you are proud of, resolve those entities, and enjoy the challenge. We are excited to see what you create.

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